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ORIGINAL ARTICLE

An analytical approach to asymmetrical wire rolling process with finite element verification

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Abstract Based on the slab method, a three-dimensional analytical solution is first proposed to calculate the rolling pressure, force, and torque in asymmetrical wire rolling where the work rolls radii, their peripheral speeds, and the interfacial frictions may be different. Since the lateral spread cannot be ignored in the wire rolling process, the plain strain condition is not applicable. Moreover, the shear and normal stresses in the vertical sides of the element are considered to be distributed linearly. The effects of process parameters such as reduction in height, rolls peripheral speed ratio, and friction factor ratio on the processes output, like pressure distribution, rolling force, and lateral spread are investigated. Moreover, the commercial finite element code, ABAQUS, is used in this study to model the asymmetrical wire rolling. The selected analytical results were compared with FEM simulation data to verify the effectiveness of the developed model. By this proposed analytical solution, it is noted that the computer expense and calculation time can be saved, and the characteristics of asymmetrical wire rolling are obtained simply and quickly.

Keywords Asymmetrical wire rolling · Slab method · Pressure distribution · Neutral point · FEM

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Nomenclature

b	Width of the wire contact area (mm)
b_{nu}, b_{nl}	Width of the wire contact area at the upper and the lower neutral points, respectively (mm)
F_u, F_l	Upper and lower rolling force, respectively (kN)
h_1, h_2	Thickness of upper and lower elements with respect to horizontal axis (mm)
h_i, h_o	Thickness of wire at entry and exit of the roll gap, respectively (mm)
h_{nu}, h_{nl}	Thickness of wire at the neutral points with respect to upper and lower rolls, respectively (mm)
Δh	Reduction in thickness (mm)
k	Shear yield stress (MPa)
l	Length of the contact area (mm)
m_u, m_l	Friction factors at the interface of wire with upper and the lower rolls, respectively
p_u, p_l	Pressure distribution at the upper and the lower rolls, respectively (MPa)
R_{eq}	Equivalent work roll radius (mm)
r	Reduction percentage in the wire thickness (%)
R_u, R_l	The upper and the lower work roll radii, respectively (mm)
T	Total rolling torque (N.m)
T_u, T_l	Rolling torques of the upper and the lower rolls, respectively (N.m)
V_{nu}, V_{nl}	Speeds of the wire at the upper and the lower neutral points, respectively (mm/s)
V_1, V_2, V_3	Different speed ratios regarding to the upper and the lower work rolls
V_u, V_l	Peripheral velocities of the upper and the lower rolls, respectively (mm/s)
x_{nu}, x_{nl}	Positions of the upper and the lower neutral points, respectively (mm)

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σ_{xu}, σ_{xl}	Normal stresses on the vertical side of the element at the top and bottom of the slab, respectively (Pa)
τ_u, τ_l	Frictional shear stresses at the interfaces of wire with the upper and the lower rolls, respectively (Pa)
$\overline{\tau}_e$	Equivalent shear stress (Pa)
α_u, α_l	Contact variable angles of the upper and lower rolls, respectively (Rad)

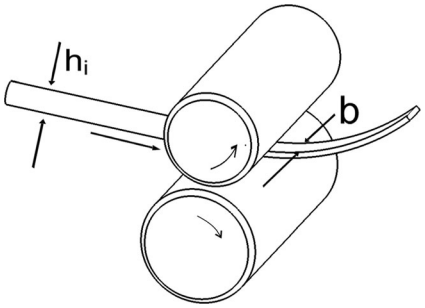


Fig. 1 Asymmetrical wire rolling process

1 Introduction

In wire rolling, wire is entered between the gap of two rolls to achieve the desired thickness reduction. The initial material for rolling is a wire of circular cross-section and final product is a flattened wire with a round edge in both sides. The wire manufactured by wire rolling process is mostly used in the production of the electrical equipment, piston rings, spring wire, bandsaw blades, and flat-wire electrode for gas metal arc welding (GMAW) [1].

In the asymmetrical wire rolling, the roll radii, peripheral speeds, and surface roughness of the work rolls may be dissimilar. These differences lead to reduce the required force and torque of the process and the total energy is saved. Moreover, the higher reduction in one pass can also be obtained. In addition to the preceding points, the outgoing curvature (see Fig. 1), which is generally regarded as a deficiency of the asymmetrical rolling process, could be considered as a constructive output of this process while it may be used to produce torsional or helical springs.

Up to now, a number of researchers have applied analytical and experimental methods to investigate the symmetrical wire rolling and asymmetrical sheet rolling processes. Kazeminezhad and Karimi Taheri [2] investigated the effects of the thickness reduction, roll speed, and wire material on the width of the contact area and lateral spread of wire by experimental study of symmetrical wire rolling process. They presented a relation to find out the width of the contact area based on the wire thickness in the deforming zone. Using an experimental investigation, Kazeminezhad and Karimi Taheri [3] evaluated the effects of the thickness reduction and roll speed on rolling force and deformation behavior of symmetrical wire rolling process. Using slab method analysis, Kazeminezhad and Karimi Taheri [4] also studied the effects of the thickness reduction, yield stress, and coefficient of friction on the positions of neutral points, rolling pressure, and force in the symmetrical wire rolling. Following their previous study, they [5] studied the deformation inhomogeneity in the flattened wire produced in the symmetrical rolling by applying finite element method using slab element.

Using the experimental and analytical formulae as well as the numerical method, the cross-sectional profile of flattened wires after the flat rolling process was investigated by Kazeminezhad et al. [6]. Moreover, Kazeminezhad and Karimi Taheri [7] predicted the effective strain fields in the flat rolled wires considering the 2D and 3D deformation. Besides, they [8] considered finite element method using slab element to investigate the creation of the macroscopic shear bands in wire flat rolling process. By means of experimental tests and finite element simulation, Parvizi et al. [9] investigated the effects of the roll speed ratio, roll diameter ratio, and reduction in height on the curvature radius and width of the contact area of a rolled wire produced by asymmetrical rolling.

Taking into consideration the slab method, the asymmetrical rolling process has been investigated by a number of researchers. Based on the slab method theory, Parvizi et al. [10] presented an analytical solution for ring rolling process with constant shear friction model ($\tau = mk$). They assumed the non-uniformity of the normal and shear stresses through the vertical side of the slabs in the deforming zone. Gudur et al. [11] presented an analytical model for the asymmetrical cold sheet rolling to calculate the friction coefficient according to the output radius of the rolled sheet. They eliminated the spread of sheet in the transverse direction and considered the constant shear stress through thickness of vertical side of the element in the analysis.

Assuming a plain strain condition, Yong et al. [12] studied the asymmetrical rolling process based on the slab method analysis. In their slab model, the normal stresses were distributed uniformly in vertical side of the element and the shear stresses were not taken into account in the calculation. Considering strain hardening and a Wanheim and Bay's friction model, Kumar and Dixit [13] performed the slab analysis of foil-rolling process. Chen et al. [14], based on the slab method, presented a theoretical model for rolling of a large cylindrical shell. They used a mixed friction model consisting of sticking and Coulomb friction to state the interface friction condition. Qwamizadeh et al. [15] developed a theoretical model with slab method to evaluate the asymmetrical rolling of bonded two-layer sheets. Zhang et al. [16] proposed an analytical method to calculate the rolling force and torque in asymmetrical sheet rolling. To verify the validity of their analytical method, they compared the analytical rolling force and

torque with the experimental and analytical results of other investigators.

In addition, a new solution for asymmetrical rolling of unbonded clad sheet was presented by Afrouz and Parvizi [17]. They considered the constant shear friction model ($\tau = mk$) as well as the non-uniformity of the shear stresses and the uniformity of the normal stresses at the slab model. Tzou and Huang [18] investigated the minimum thickness for asymmetrical cold-and-hot PV rolling by assuming constant shear friction model ($\tau = mk$). Furthermore, Tzou [19] developed an analytical model to study the relationship between friction factor and friction coefficient in the asymmetrical sheet rolling process.

Although finite element method may be used to simulate the metal forming processes with an acceptable accuracy, but it however involves too much computation time and convergence problems due to the existence of complicated circumstances such as boundary conditions, contact conditions, etc. In addition to the theoretical and experimental analysis, FEM is also utilized by some researchers to analyze the symmetrical wire flat rolling process. Carlsson [1] simulated the wire flat rolling process for evaluating the pressure distribution. He concluded that the maximum pressure happens at the entry of the deforming zone. Iankov [20] estimated the effects of the friction and back and front tension on the contact pressure distribution and lateral spread in the symmetrical wire flat rolling process. Vallellano et al. [21] investigated the contact stress distribution and residual stress in the rolled wire by 3D finite element method. They found that there is a maximum contact pressure in the roll entry zone due to the strong local inhomogeneity of deformation.

Comparing the wire and sheet rolling processes, a key difference is apparent. In point of fact, supposing a plane strain condition in the sheet rolling process is expected, consequently the lateral spread of sheet is negligible. On the other hand, the lateral spread cannot be ignored in the wire rolling process. Thus, in addition to wire height variation, the variation of wire width should also be considered in the slab analysis of this process. This complex condition causes that the slab analysis of wire rolling process has been remained so limited. In other words, there is hardly any slab analysis of wire rolling process except that given by Kazeminezhad and Karimi Taheri [4] which was presented for the symmetrical process type. Consequently, no research has been reported for the asymmetrical wire rolling process until now.

The present work is motivated by the lack of analytical solutions for the asymmetrical wire rolling process. Using a slab method and FEM simulation, the asymmetrical wire rolling process is investigated for the first time in this paper. As a result of this solution, the rolling pressure distribution, rolling force, and torque can be obtained simply and quickly. Using ABAQUS explicit, the asymmetrical wire rolling process is simulated in every aspect. The analytical results are shown to be in good agreement with those from finite element simulation. Moreover, effects of the reduction in height, roll speed ratio, roll diameter ratio, and friction factor ratio on the rolling force and rolling torque are discussed thoroughly.

2 Mathematical model

In the development of the slab method analysis, the following assumptions are considered:

1. A constant shear stress model ($\tau = mk$) is used to express interface friction between rolls and wire. Therefore, the friction factor (m) between the rolls and wire is assumed to be constant along the contact length.
2. Both rolls are supposed to be driver.
3. The elastic deformation is negligible.
4. Variations of the normal and shear stresses on the vertical sides of each slab are linear and the material spread in transverse direction is considered.
5. The work rolls are rigid and the wire being rolled is rigid perfectly plastic with no strain-hardening effect.
6. The contact length between wire and the upper and the lower rolls in the deformation zone are equal.

Figure 2a shows the asymmetrical wire rolling geometry. Because of the asymmetrical nature of the process, the neutral points, where the velocity of wire becomes equal to the peripheral velocities of the upper and the lower work rolls, most likely not located in a same vertical plane. Therefore, a cross shear zone is formed between backward and forward slip zones. Consequently, in general case, the plastic deformation zone at the roll gap can be divided into three different zones.

In zone I ($0 \le x \le x_{nl}$), peripheral velocity of both rolls are more than the speed of the wire and the frictional stresses on the upper and the lower sides of slab are in the forward direction. The horizontal distance from the entry plane in the roll gap is taken as the x-axis and the origin is its intersection with the wire mid plane. In zone II ($x_{nl} \le x \le x_{nu}$) in which the speed of wire is more than the peripheral speed of the lower roll and less than the speed of the upper roll, the direction of the frictional stresses on the upper and lower surfaces of the slab are opposite. Zone III ($x_{nu} \le x \le l$) is similar to zone I, but the direction of the frictional stresses on the upper and the lower surfaces of slab are backward while the velocity of the wire is more than those of the both rolls.

In contrast to the typical assumption of plane strain condition in the asymmetrical sheet rolling process, the lateral spread of wire in asymmetrical wire rolling process is noticeable. Hence, the width of wire in the contact area is one of the important parameters that should be considered in the wire rolling process. For this reason, the slab that is considered in this study is a three-dimensional element (Fig. 2b) in which

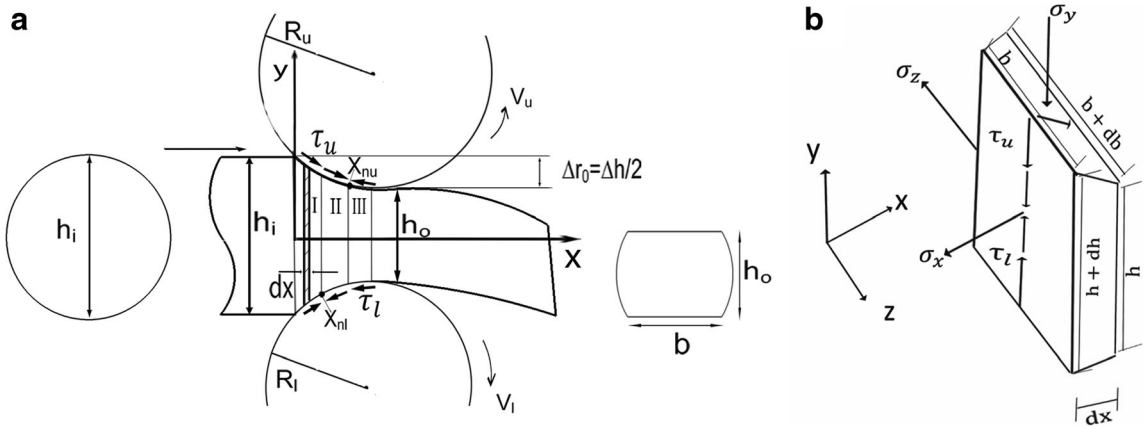


Fig. 2 **a** Asymmetrical wire rolling geometry; **b** three-dimensional view of element in deforming zone

the variations in the z direction have also been taken into account. Moreover, due to the different roll diameter of the upper and lower rolls, their contact angles with the wire, α_u and α_l, are different. The normal and shear stresses on the vertical sides of the differential elements are assumed to be linearly distributed. The stress fields acting on different vertical elements at the deforming zone are shown in Fig. 3.

The length of the contact area in the asymmetrical wire rolling is small as compared with the circumferences of the rolls and can be obtained from the following relation [17]:

$$l = \sqrt{R_{eq}(h_i - h_0)} \quad R_{eq} = 2R_u R_l \Big/ (R_u + R_l) \tag{1}$$

where, h_i, h_0 are the initial and final thicknesses of the wire. Subscripts u and l indicate the quantities at the upper and the lower rolls, respectively. From simple geometry [17]:

$$h_1 = \frac{h_0}{2} + \frac{x^2}{2R_u} \quad h_2 = \frac{h_0}{2} + \frac{x^2}{2R_l} \tag{2}$$

where h_1 and h_2 are height of upper and lower parts of element with respect to horizontal axis, respectively. Therefore, element total height (thickness) can be written as follows:

$$h = h_1 + h_2 = h_0 + \frac{x^2}{R_{eq}} \tag{3}$$

where R_{eq} is equivalent work roll radius.

2.1 Zone I ($0 \le x \le x_{nl}$)

Applying horizontal and vertical force equilibrium for an element in zone I (Fig. 3a), the following relationships are obtained, respectively:

$$\begin{aligned} &\frac{(h\ \sigma_{xu}\ \ db)}{2} + \frac{dh\ \sigma_{xu}\ b}{2} + \frac{h\ d\sigma_{xu}\ b}{2} + \frac{(h\ \sigma_{xl}\ db)}{2} \\ &\quad + \frac{dh\ \sigma_{xl}\ b}{2} + \frac{h\ d\sigma_{xl}\ b}{2} p_u \tan\alpha_u\ \ b\ dx \\ &\quad + p_l \tan\alpha_l\ b\ dx - b\ dx\ (\tau_e) = 0 \end{aligned} \tag{4}$$

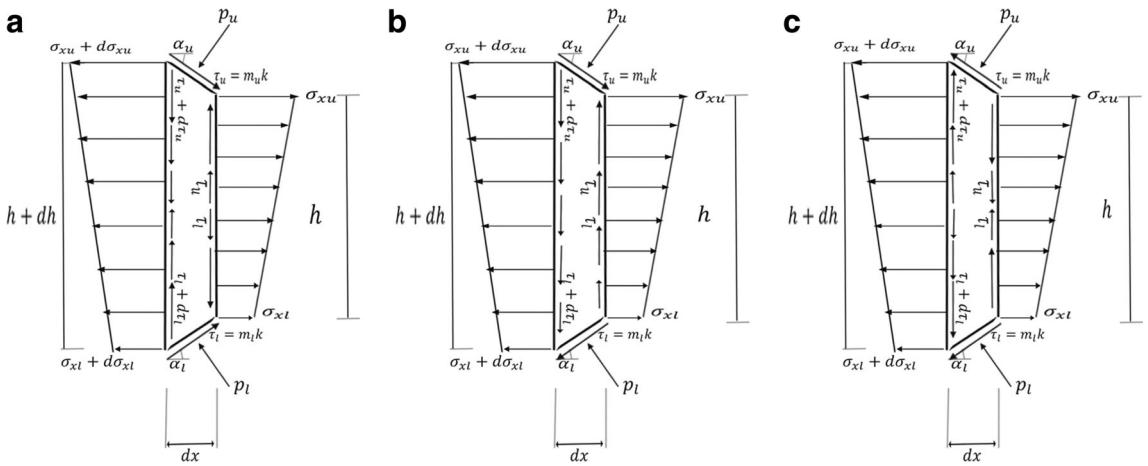


Fig. 3 Stress field acting on a differential vertical element at **a** zone I; **b** zone II; and **c** zone III

$$p_l\ b\ dx+\tau_l\ b\ dx\ \tan\alpha_l-\left(\frac{\tau_l\ b\ h}{4}\right)-(\tau_l+d\tau_l)\left(\frac{h+dh}{2}\right)\left(\frac{b+db}{2}\right)-p_u\ b\ dx-\tau_u\ b\ dx\ \tan\alpha_u+\left(\frac{\tau_u\ b\ h}{4}\right)-(\tau_u+d\tau_u)\left(\frac{h+dh}{2}\right)\left(\frac{b+db}{2}\right)=0 \tag{5}$$

From Eq. (5), the pressure at the lower roll can be obtained from the upper roll pressure as:

$$p_l=p_u+\tau_u\ \tan\alpha_u+\left(\frac{\tau_u\ h\ db}{4\ b\ dx}\right)+\left(\frac{\tau_u\ dh}{4dx}\right)-\tau_l\tan\alpha_l-\left(\frac{\tau_l\ h\ db}{4\ b\ dx}\right)-\left(\frac{\tau_l\ dh}{4\ dx}\right) \tag{6}$$

where σ_{xu} and σ_{xl} are normal stresses on vertical side of element at the top and bottom of the slab, α_u and α_l are the contact angle of the upper and lower rolls, p_u and p_l are upper and lower rolls pressures, τ_e is equivalent shear stress that can be stated as follows:

$$\tau_e=\tau_u+\tau_l \tag{7}$$

Supposing that the shear stress in the z direction is negligible, the normal stress on that direction can be considered as a principal stress. It is also assumed that the quantity of principle stress in z direction is between two other principle stresses [4], so the Tresca yield criterion for the elements on a vertical slab takes the following form [9]:

$$(\sigma_x)-(\sigma_y)=2k\sqrt{1-m^2} \tag{8}$$

Therefore, the yield criteria for elements located at the top and the bottom of each slab in zone I have the following forms, respectively:

$$\sigma_{xu}=2k\sqrt{1-m_u^2}-p_u \tag{9}$$

$$\sigma_{xl}=\sqrt{1-m_l^2}-p_l \tag{10}$$

From Eqs. (9) and (10), it can be concluded that:

$$d\sigma_{xu}=-dp_u \tag{11}$$

$$d\sigma_{xl}=-dp_l \tag{12}$$

The width of the contact area in the symmetrical wire flat rolling process can be obtained as follows [2]:

$$b=\sqrt{2\Delta h\ h_i} \tag{13}$$

where Δh is the reduction in height of wire. Comparing the results from FEM simulation of asymmetrical wire rolling process with those from Eq. (13), it is concluded that this equation can also be applied for the asymmetrical wire rolling process. From Fig. 2, since $\Delta r_0=\frac{\Delta h}{2}$, the following relation can also be obtained:

$$b=\sqrt{4h_i\Delta r}\quad,\quad \Delta r=\frac{\Delta h}{2}-\frac{x^2}{2\ R_e} \tag{14}$$

Applying Eqs. (7), (9)–(12) into Eq. (6) and substitute the subsequent result into Eq. (5), the differential equation for the upper roll pressure is obtained as follows:

$$\frac{dp_{u(\mathbf{I})}}{dx}=\frac{x}{2\ R_e\ \Delta r_0-x^2}\left(\begin{array}{c}k\sqrt{1-m_u^2}+\sqrt{1-m_l^2}-p_{u(\mathbf{I})}\\+\ \tau_l\frac{x}{2R_e}+\tau_u\frac{x}{4R_e}-\tau_l\frac{x}{4R_e}\end{array}\right)+\frac{2x}{R_e h_0+x^2}\left(k\sqrt{1-m_u^2}-k\sqrt{1-m_l^2}-p_{u(\mathbf{I})}-\tau_u\frac{x}{2R_u}+\tau_l\frac{x}{2R_l}+\tau_u\frac{x}{4R_l}-\tau_l\frac{x}{4R_l}\right)+\frac{2R_e}{R_e h_0}\left(2p_u\frac{x}{R_u}+\tau_u\frac{x^2}{2R_u}-\tau_l\frac{x^2}{2R_l}-\tau_u-\tau_l\right) \tag{15}$$

2.2 Zone II ($x_{nl}\le x\le x_{nu}$)

Carrying out the similar calculation given in Section 2.1, the differential equation for the upper roll pressure in the element located in zone II (Fig. 3b) is obtained as follows:

$$\frac{dp_{u(\mathbf{II})}}{dx}=\frac{x}{2\ R_e\ \Delta r_0-x^2}\left(\begin{array}{c}k\sqrt{1-m_u^2}+\sqrt{1-m_l^2}-p_{u(\mathbf{II})}\\-\tau_l\frac{x}{2R_e}+\tau_u\frac{x}{4R_e}+\tau_l\frac{x}{4R_e}\end{array}\right)+\frac{2x}{R_e\ h_0+x^2}\left(k\sqrt{1-m_u^2}-k\sqrt{1-m_l^2}-p_{u(\mathbf{II})}-\tau_u\ \frac{x}{2R_u}-\tau_l\ \frac{x}{2R_l}+\tau_u\frac{x}{4R_l}+\tau_l\frac{x}{4R_l}\right)+\frac{2R_e}{R_e h_0}\left(2p_u\frac{x}{R_u}+\tau_u\frac{x^2}{2R_u}+\tau_l\frac{x^2}{2R_l}-\tau_u+\tau_l\right) \tag{16}$$

2.3 Zone III ($x_{nu} \le x \le L$)

Similar to preceding parts, the differential equation for upper roll pressure in zone III (Fig. 3c) is also obtained as follows:

$$\frac{dp_{u(III)}}{dx} = \frac{x}{2\ R_e \Delta r_0 - x^2}\left(\begin{array}{c} k\sqrt{1-m_u^2} + \sqrt{1-m_l^2} - p_{u(III)} \\ -\tau_l \frac{x}{2R_e} - \tau_u \frac{x}{4R_e} + \tau_l \frac{x}{4R_e} \end{array}\right) + \frac{2x}{R_e\ h_0 + x^2}\left(k\sqrt{1-m_u^2} - k\sqrt{1-m_l^2} - p_{u(III)} + \tau_u \frac{x}{2R_u} - \tau_l \frac{x}{2R_l} - \tau_u \frac{x}{4R_l} + \tau_l \frac{x}{4R_l}\right) + \frac{2R_e}{R_e h_0}\left(2p_u \frac{x}{R_u} - \tau_u \frac{x^2}{2R_u} + \tau_l \frac{x^2}{2R_l} + \tau_u + \tau_l\right) \tag{17}$$

From rule of constant volume rate, the following equation can be applied in the neutral points at the upper and the lower surfaces [12]:

$$V_{nl} h_{nl} b_{nl} = V_{nu} h_{nu} b_{nu} \tag{18}$$

where subscripts nl and nu are related to the neutral points at the lower and upper surfaces, respectively. Furthermore, h_{nl}, h_{nu}, b_{nl}, and b_{nu} can be substituted from Eqs. (3) and (14), correspondingly. Taking into consideration Eq. (18) and by means of Rung-Kutta method, three differential Eqs. (15), (16), and (17) can be solved numerically. Having the pressure distribution at the upper roll (p_u), the pressure distribution at the lower roll (p_l) can also be obtained easily by means of Eq. (6).

3 Rolling force

Assuming a small contact angle of the rolls and that of the wire, the rolling force at the interface of wire with the upper roll can be obtained as follows:

$$F = F_I + F_{II} + F_{III} = \int_o^{x_{nl}} \left(p_{u(I)} b\right) dx + \int_{x_{nl}}^{x_{nu}} \left(p_{u(II)} b\right) dx + \int_{x_{nu}}^{l} \left(p_{u(III)} b\right) dx \tag{19}$$

4 Rolling torque

The rolling torques, T_u and T_l, exerted by the wire on the upper and lower rolls, respectively, can be calculated by integrating the moment of shear stress around the roll axis as follows:

$$T_u = R_u\left[\left(\int_0^{x_{nl}} m_u k\ b\ dx\right) + \left(\int_{x_{nl}}^{x_{nu}} m_u k\ b\ dx\right) - \left(\int_{x_{nu}}^{l} m_u k\ b\ dx\right)\right] \tag{20}$$

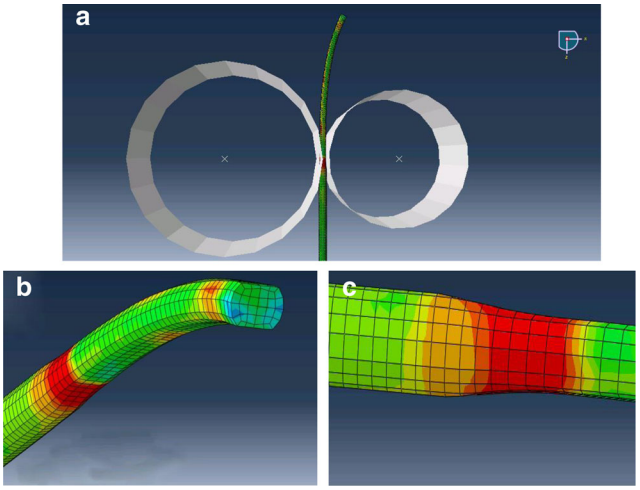


Fig. 4 **a** 3D geometric model used in simulation; **b** meshing in FEM; **c** stress contour at the deforming zone

$$T_l = R_l\left[\left(\int_0^{x_{nl}} m_l k\ b\ dx\right) - \left(\int_{x_{nl}}^{x_{nu}} m_l k\ b\ dx\right) + \left(\int_{x_{nu}}^{l} m_l k\ b\ dx\right)\right] \tag{21}$$

And the total required rolling torque is

$$T = T_u + T_l \tag{22}$$

5 FEM simulation

To evaluate the accuracy of the analytical results, the asymmetrical wire rolling process is simulated using ABAQUS software as shown in Fig. 4a. The dynamic-explicit analysis was assumed to simulate the nonlinear behavior of the rolling process. The mesh sensitivity assessment was carried out to check and approve the correct number of the elements. Mass scaling technique was also assumed to improve the convergence of the simulation. 3D analytic rigid shell and C3D8R 3D elements were considered to make a mesh model of the rolls and the wire, respectively. Based on the initial diameter of wire, 32 or 48 elements in the rolled wire cross section with the arrangement demonstrated in Fig. 4b were considered.

Two rolls are only allowed to rotate around the y-axis and constrained in the x and z directions. Furthermore, the wire is also constrained in the y direction. As a result of the friction,

Table 1 Material properties used in the FEM and analytical studies

Material	Shear yield stress (MPa)	Modulus of elasticity (GPa)	Poisson's ratio	Density (kg/m^3)
UNS C23000	125	115	0.31	8940
UNS C46400	98.1	115	0.31	8940
UNS C34200	165	115	0.28	8920

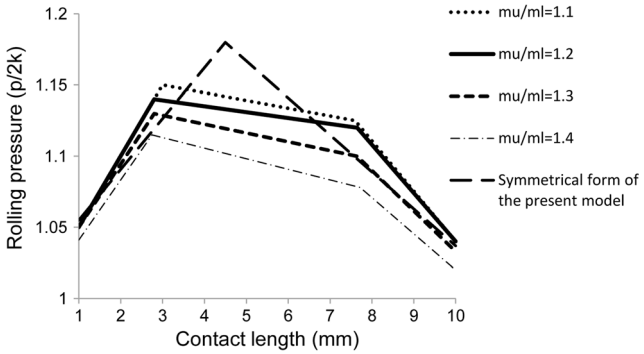


Fig. 5 The comparison between the symmetrical and asymmetrical wire rolling processes for specific pressure distribution along the contact length $R_u = R_l = 75$ mm, $\frac{V_u}{V_l} = 1.1$, $V_l = 50\frac{mm}{s}$, $h_i = 5.5$ mm, $\Delta h = 0.9$ mm, $m_u + m_l = 0.7, k = 125$ MPa

the wire entered the rolls gap and flattened wire was obtained at the exit of the deformation zone. Figure 4c shows the stress contour at the deforming zone. The material properties considered in the FE simulation are shown in Table 1 which also be applied in the analytical studies. Moreover, details of process parameters are given in the description of Figs. 5, 6, 7, 8, 9, 10, and 11.

6 Result and discussions

The comparison between the symmetrical and asymmetrical wire rolling processes for specific pressure distribution along the contact length is given in Fig. 5. The roll friction factor ratio was considered as a variable parameter to investigate the different asymmetrical conditions of the process. Unlike to the symmetrical process which has one peak point in its curve, the asymmetrical curves include two peak points. These points in each curve demonstrate the positions of the neutral points.

In comparison to the symmetrical process, it is apparent based on this figure that the rolling pressure can be decreased significantly by increasing the asymmetrical condition of the process. In other words, as friction factor ratio increases, the rolling pressure decreases and the length of the contact area in zone II increases slightly. In addition, the neutral point at the interface of the wire with the roll having lower peripheral speed moves towards the entry of the deformation zone. While at the other interface, the neutral point has tendency to move towards the exit. To have a logical comparison in this case study, it is assumed that the summation of the friction factors at the upper and lower rolls is constant, i.e., $m_u+m_l=0.7$. Therefore, by increasing the upper friction factor, the lower friction factor should be decreased to preserve the constant summation. According to this figure, the more asymmetrical nature of the process will result into the lower pressure and required force.

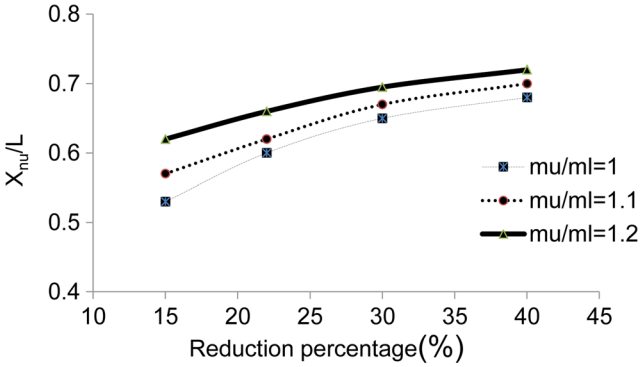


Fig. 7 Effect of the reduction percentage on the position of the neutral point $R_u = R_l = 75$ mm, $\frac{V_u}{V_l} = 1.1$, $V_l = 55$mm/s, $h_i = 5.5$ mm, $m_u + m_l = 0.7$

Figure 6 illustrates the effect of the reduction in height on specific rolling pressure distribution. As it is seen, for higher reductions in height, the higher rolling pressures are required. It is also apparent that variation in reduction in height leads to have changes in the position of the neutral points. For all reductions, it seems that the position of the first neutral point is located at almost 25 % of the contact length. But, the positions of the second neutral points are irregular and do not have a specific relation with the contact length.

Figure 7 demonstrates the effect of the reduction percentage on the positions of the neutral point at the interface of the

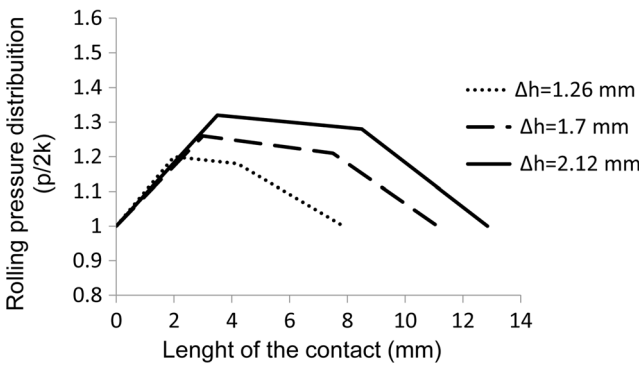


Fig. 6 Specific pressure distribution for different reductions in height $R_u = R_l = 75$ mm, $\frac{V_u}{V_l} = 1.1$, $V_l = 55\frac{mm}{s}$, $h_i = 5.5$ mm, $m_u = m_l = 0.3$, $k = 125$ MPa

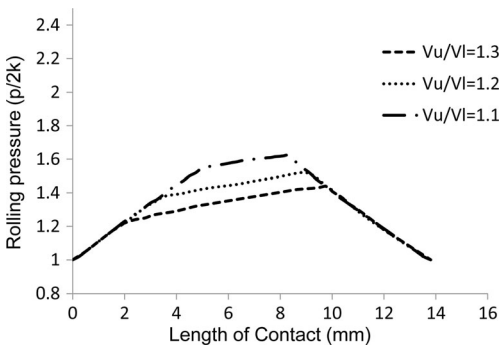


Fig. 8 Specific rolling pressure distribution for different roll peripheral speed ratios h_i=3.2 mm, R_l=R_u=200 mm, V_l=40mm/s, m_l=m_u=0.5, r=10 %, k=98.1 MPa

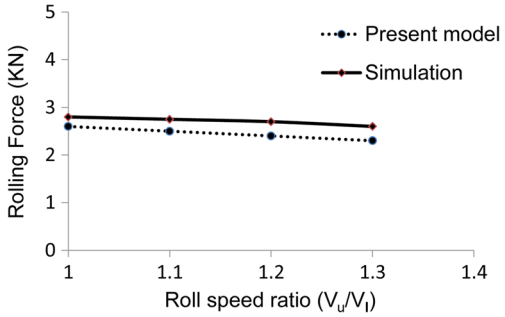


Fig. 9 Effect of the roll peripheral speed ratios on the rolling force and comparison with FEM simulation h_i=3.2 mm, R_l=R_u=200 mm, V_l= 40 mm/s, m_l=m_u=0.5, r=10 %, k=98.1 MPa

wire with the upper roll. Based on this figure, by increasing the reduction, the value of the ($\frac{x_{nu}}{L}$) rises and the upper neutral point moves toward the exit. Moreover, considering a constant reduction, the position of the neutral point at the upper roll gets closer to the exit for the higher friction factor ratio.

The specific rolling pressure distributions along the contact length for different roll peripheral speed ratio are shown in Fig. 8. It is seen that when the roll peripheral speed ratio increases, the rolling pressure decreases and the neutral points get farther from each other. Therefore, the length of the cross-section zone increases. The lower rolling pressure and force are the consequences of having higher roll peripheral speed ratio which is related to the further asymmetrical behavior of the process.

The effects of the roll peripheral speed ratios on the rolling force are shown in Fig. 9 where the results are compared with those from FEM simulation. It is seen that even though the rolling force calculated by theoretical formula is less than those ones from simulation, there is a good agreement between the results. In other words, the maximum difference between two results is less than 10 %. Furthermore, it is obvious that the rolling force drops to some extent by increasing the peripheral speed ratios.

Figure 10 shows the variation of rolling torque with respect to different roll diameter ratios. The lower roll diameter is considered to be constant; therefore, the upper roll diameter is changed to afford the desired roll diameter ratios. For both

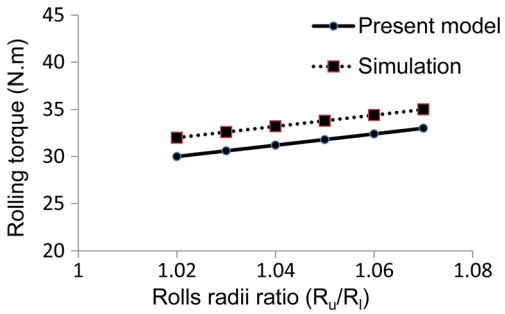


Fig. 10 Variation of rolling torque with rolls radii ratios $h_i = 3.2$ mm, $\frac{V_l}{V_u} = 1.05$, $R_l = 200$ mm, $V_l = 40$ mm/s, $m_u = m_l = 0.35$, $\Delta h = 10$ %, $k = 98.1$ MPa

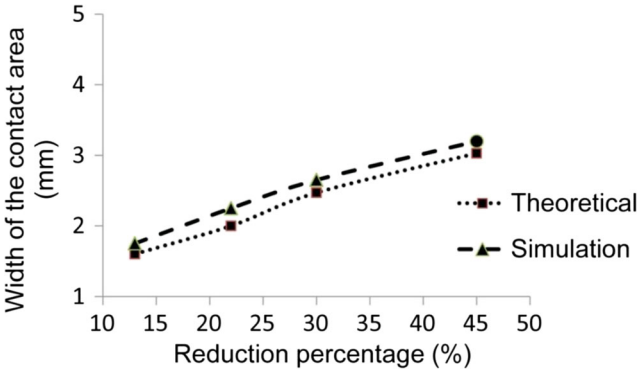


Fig. 11 Variation of width of the contact area with reduction percentage $h_0 = 3.2$ mm, $V_u = V_l = 50$ mm/s, $m_u = m_l = 0.4$, $k = 165$ MPa, $\frac{R_u}{R_l} = 1.4$

theoretical and simulation conditions, the rolling torque increases as the roll diameter ratio rises.

The effect of the reduction percentage on width of the contact area is demonstrated in Fig. 11 where the theoretical results are compared with the simulation ones. It is seen that the values from theoretical results are in good conformity with those from FE simulation. Therefore, the assumption of applying the theoretical Eq. (13), which was generated for a symmetrical process, to the asymmetrical process is acceptable. Furthermore, it is seen that by increasing the reduction percentage, the width of the contact area increases too.

7 Conclusion

In this research, for the first time, the slab analysis as well as the FEM simulation for the asymmetrical wire rolling process is presented. According to the analytical solution, the rolling pressure distribution, rolling force, and torque can be obtained promptly. It is found that the rolling pressure and force can be decreased by increasing the asymmetrical nature of the process through getting higher friction factor ratio or rolls peripheral speed ratio. It is also seen that the length of cross shear zone increases in these cases. On the contrary, the higher reduction percentage will require higher pressure and force in the asymmetrical wire rolling process. In all cases, the positions of the neutral points have a significant effect on the process outputs. The simulation results reveal that increasing the rolls diameter ratio has a negligible effect on the width of the contact area. Having good conformities between simulation results for width of contact area with those from the theoretical formula related to symmetrical wire flat rolling proves that Eq. (13) can also be applied to the asymmetrical wire rolling process. The slab method code incorporating shear and normal stresses in the element will prove an enhanced alternative to rigorous finite element analysis, when fast computational speed is also a requirement.

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References

1. Carlsson B (1998) The contact pressure distribution in flat rolling of wire. J Mater Process Technol 15:1–6
2. Kazeminezhad M, Karimi Taheri A (2005) A theoretical and experimental investigation on wire flat rolling process using deformation pattern. J Mater Des 26:99–103
3. Kazeminezhad M, Karimi Taheri A (2005) An experimental investigation on the deformation behavior during wire flat rolling process. J Mater Process Technol 160:313–320
4. Kazeminezhad M, Karimi Taheri A (2006) Calculation of rolling pressure distribution and force in wire flat rolling process. J Mater Process Technol 171:253–258
5. Kazeminezhad M, Karimi Taheri A (2007) Deformation inhomogeneity in flattened copper wire. J Mater Des 28:2047–2053
6. Kazeminezhad M, Karimi Taheri A, Tieu AK (2008) A study on the cross-section profile of flat rolled wire. J Mater Process Technol 200:325–330
7. Kazeminezhad M, Karimi Taheri A (2008) The effect of 3D and 2D on flattened wires. J Mater Process Technol 202:553–558
8. Kazeminezhad M, Karimi Taheri A (2006) The prediction of the macroscopic shear bands in flat rolled wire using the finite and slab element method. Mater Lett 60:3265–3268
9. Parvizi A, Pasoodeh B, Abrinia K, Akbari H (2015) Analysis of curvature and width of the contact area in asymmetrical rolling of wire. J Manuf Process. doi:10.1016/j.jmapro.2015.07.004
10. Parvizi A, Abrinia K, Salimi M (2011) Slab analysis of ring rolling assuming constant shear friction. J Mater Eng Perform 20:1505–1511
11. Gudur PP, Salunkhe MA, Dixit US (2008) A theoretical study on the application of asymmetric rolling for the estimation of friction. Int J Mech Sci 50:315–327
12. Yong T, Yan-hui G, Wang ZD, Wang GD (2009) Analysis of rolling pressure in asymmetrical rolling process by slab method. J Iron Steel Res Int 16(4):22–26
13. Kumar D, Dixit US (2006) A slab method study of strain hardening and friction effects in cold foil rolling process. J Mater Process Technol 171:331–340
14. Chen S, Liu H, Peng Y, Sun J (2014) Slab analysis of large cylindrical shell rolling considering mixed friction. Int J Mech Sci Technol 28:4753–4760
15. Qwamizadeh M, Kadkhodaei M, Salimi M (2014) Asymmetrical rolling analysis of bonded two-layer sheets and evaluation of outgoing curvature. Int J Adv Manuf Technol 73:521–533
16. Zhang SH, Zhao DW, Gao CR, Wang GD (2012) Analysis of asymmetrical sheet rolling by slab method. Int J Mech Sci 65: 168–176
17. Afrouz F, Parvizi A (2015) An analytical model of asymmetric rolling of unbounded clad sheets with shear effects. J Manuf Process. doi:10.1016/j.jmapro.2015.08.007
18. Tzou GY, Huang MN (2001) Study on minimum thickness for asymmetrical hot-and-cold PV rolling of sheet considering constant shear friction. J Mater Process Technol 119(3):229–233
19. Tzou GY (1999) Relationship between frictional coefficient and frictional factor in asymmetrical sheet rolling. J Mater Process Technol 86:271–277
20. Iankov R (2003) Finite element simulation of profile rolling wire. J Mater Process Technol 142:355–361
21. Vallellano C, Cabanillas PA, Garcia-Lomas FJ (2008) Analysis of deformation and stress in flat rolling of wire. J Mater Process Technol 195:63–71